

Simple Authentication with PHP and MySQL

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Overview and Requirements

- This lesson describes how to do very simple user authentication. This method is for learning purposes – it is not secure and would not be implemented like this in an enterprise or production environment.
- Requirements:
 - A web server running PHP, and a MySQL server
 - A login form with a target page
 - A MySQL database with a table that stores usernames and passwords

Additional Resources

- W3Schools:
 - [PHP form handlingLinks to an external site.](#) - use the \$_POST superglobal to move data from one page to another
 - [PHP MySQL Database connectionLinks to an external site.](#) - review the MySQLi example, and see it implemented in the example files
- GitHub:
 - Simple user authentication scripts are available on the instructor's GitHub and in Canvas. You may download these before you begin.

Database

- For this exercise, you should have an existing database with a table of usernames and passwords.
- We will run a SELECT query against the table for one username and password.
- Screenshot of table:

username	password
gggeorge	password
profmayhem	password
thejames	password
tjones	password

We will log in with this
username and password

MySQL query

- Purpose: to select a matching username/password pair from the database.

- Query:

```
SELECT * from auth WHERE Username = 'profmayhem' AND Password = 'password';
```

- To review the SELECT query syntax, please visit:
 - https://www.w3schools.com/mysql/mysql_select.asp
 - https://www.w3schools.com/mysql/mysql_where.asp
 - https://www.w3schools.com/mysql/mysql_and_or.asp

The Login Form (PHP)

- You should already have a login form from the previous assignments. The form and the HTML should look something like this:

Username:

Password:

Submit

```
<form action="phpformtarget1.php" method="POST">
  <label for="username">Username:</label><br>
  <input type="text" name="username"><br>
  <label for="password">Password:</label><br>
  <input type="text" name="password"><br><br>
  <input type="submit" value="Submit">
</form>
```

PHP \$_POST SuperGlobal values

- The name attributes of the text boxes are copied into the \$_POST superglobal array. You can review PHP form handling here:
 - https://www.w3schools.com/php/php_forms.asp
- The names are highlighted in the HTML on the previous slide. They are 'username' and 'password'.
- You should have used the \$_POST superglobal to display data on the target page in your PHP Prototype assignment.

Login Form Target Page

- The “target” page contains PHP scripts to process the form data.
- In the example provided, the target page does the following:
 1. Connects to the database and establishes a connection object (this is the variable `$conn` in the example);
 2. Retrieves the form data from the `$_POST` superglobal and assigns the values to variables;
 3. Uses the variables to construct a `SELECT` query;
 4. Runs the query; and
 5. Evaluates the dataset returned by the query. If the number of rows is greater than zero, it assumes the username and password were correct.

Constructing the Query

- Here we will construct an SQL query with the form values using PHP
- This can be the trickiest part, due to the different quotes that are required!
- The Query:

```
SELECT * from auth WHERE Username = 'profmayhem' AND Password = 'password';
```

- The PHP Statement:

`$sql = "SELECT * from auth WHERE Username = ' . $username . ' AND Password = ' . $password . '";`

The text in the double-quotes is " a string literal. Note we have the first single-quote of the SQL statement included here.

`$username` and `$password` are PHP variables

The `.` does concatenation

Another string literal, with the single-quotes included

Etc. See the example for the rest

Download the examples

- Please download the examples as a .zip file from Canvas, or from GitHub. You should be able to make just a few changes to implement simple user authentication.